

Danh mục: medication

Type 2 diabetes in adults: management

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NICE guideline

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How to introduce medicines

Recommendations in this section that cover dipeptidyl peptidase-4 (DPP-4) inhibitors, glucagonlike peptide-1 (GLP-1) receptor agonists, sulfonylureas and sodium–glucose cotransporter-2 (SGLT-2) inhibitors refer to each of these groups of medicines at class level.

For GLP-1 receptor agonists, at the time of publication (February 2026) this only includes liraglutide, dulaglutide, and semaglutide. For subcutaneous semaglutide (Ozempic), this only includes doses up to 1 mg once a week.

1.20 Introducing medicines in a stepwise manner

1.20.1

Introduce medicines in a stepwise manner, checking for tolerability and effectiveness of each medicine.

[2015]

1.20.2

When an adult with type 2 diabetes starts initial therapy with metformin and one or more other medicines:

introduce the medicines one at a time, starting with metformin and checking tolerability

if using an SGLT-2 inhibitor, start this as soon as metformin is at the

maximum tolerated dose

if using a GLP-1 receptor agonist or tirzepatide, start this as soon as the SGLT-2 inhibitor is at the maximum tolerated dose.

For guidance on how to prevent any negative impact on the person's eyes from starting blood glucose lowering treatment, see the

section on effects of a rapid reduction in HbA1c in NICE's guideline on diabetic retinopathy

[2022, amended 2026]

1.21 Preventing diabetic ketoacidosis when taking SGLT-2 inhibitors

1.21.1

Before starting an SGLT-2 inhibitor, check whether the person may be at increased risk of diabetic ketoacidosis (DKA), for example if they:

have had a previous episode of DKA

are unwell with intercurrent illness

are at risk of dehydration or volume depletion

are following a very low carbohydrate or ketogenic diet.

[2022, updated 2026]

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1.21.2

Address modifiable risks of DKA before starting an SGLT-2 inhibitor. For example, people who are following a very low carbohydrate or ketogenic diet may need to delay treatment until they have changed their diet.

[2022]

1.21.3

Advise adults with type 2 diabetes who are taking an SGLT-2 inhibitor that a very low carbohydrate or ketogenic diet would increase their risk of DKA and so:

they should speak with their healthcare professional before starting such a diet, and their SGLT-2 inhibitor treatment may need to be suspended for the duration of the diet.

[2022]

Why the committee made these recommendations

When starting first-line therapy with metformin and other medicines, the committee noted the importance of introducing the medicines sequentially. This enables any side effects and intolerances from the first medicine to be identified before the second is introduced. In line with current practice, the committee recommended starting with metformin and then adding the SGLT-2 inhibitor without delay once metformin

tolerability at the highest achievable dose

is established, to avoid people remaining on metformin alone for prolonged periods.

The committee agreed that sudden reductions in HbA1c can increase the risk of diabetic retinopathy and ischaemic maculopathy, especially with GLP-1 receptor agonists and tirzepatide. Given this safety concern, they highlighted relevant recommendations in NICE's guideline on diabetic retinopathy for further guidance for anyone starting these medicines.

Preventing diabetic ketoacidosis when taking SGLT-2 inhibitors

The committee noted some particularly important safety considerations to take into account before an adult with type 2 diabetes starts on an SGLT-2 inhibitor. In the committee's experience there have been multiple instances of avoidable diabetes ketoacidosis (DKA) resulting in hospital admission. The committee highlighted some factors that might put someone at higher risk of DKA, but the list is not intended to be exhaustive. Addressing modifiable risk factors before starting an SGLT-2 inhibitor could reduce the risk of DKA and make the medicine safer for the person with type 2 diabetes. The committee agreed that taking these factors into account was more important than providing a specific HbA1c threshold from which to avoid prescribing SGLT-2 inhibitors.

The committee was aware that adults with type 2 diabetes who are living with overweight or obesity may wish to try a ketogenic diet to reverse or reduce the severity of their diabetes or induce remission. However, the committee agreed, based on their experience, that there may be an increased risk of DKA associated with SGLT-2 inhibitors and such diets. It is important to tell people about these risks and to advise them to discuss any planned change to a very low carbohydrate or ketogenic diet with their healthcare professional first.

Full details of the evidence and the committee's discussion are in

evidence review B: pharmacological therapies with cardiovascular and other benefits in people with type 2 diabetes

How the recommendations might affect practice

The recommendations are not expected to significantly increase consultation time or be a change in practice because these should already form part of the prescribing process. Checking that the person is not at increased risk of DKA when they are prescribed an SGLT-2 inhibitor should help reduce the number of people who experience DKA and thereby reduce unnecessary hospital admissions.